Constraint Based Phonology

Dr. Josef Fruehwald

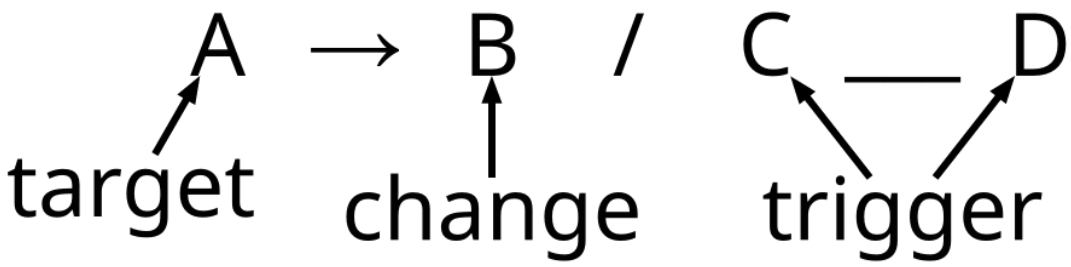
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source(here::here("_defaults.R"))
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Loading required package: sysfonts
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Loading required package: showtextdb
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Rules Recap

The anatomy of a rule



Rules Recap

Voiceless stops get deleted following a nasal.

T DELETE: $\begin{bmatrix} +\text{ant} \\ -\text{voice} \\ -\text{cont} \end{bmatrix} \rightarrow \varnothing / [+\text{nasal}]____$

PALATALIZE: $\begin{bmatrix} \text{COR} \\ -\text{dist} \end{bmatrix} \rightarrow [-\text{ant}] / ____ \begin{bmatrix} \text{DORS} \\ -\text{cons} \\ -\text{syll} \end{bmatrix}$

Rules Recap

Derivations

	don't	*don't you*
UR	doʊnt	doʊntju
PALATALIZE	—	doʊntʃju
T DEL	doʊn	—
SR	doʊn	doʊntʃju

What about phonotactics?

Remember this impossible vs impossible word of English?

blɪk *tlɪk

What about phonotactics?

To avoid *tl, people resort to multiple strategies

Chipotle

- Input: tʃɪ.poʊ.tle

- Outputs
  - /e/ deletion: tʃɪ.pa.ɾ̩
  - Metathesis: tʃɪ.poʊɬ.te
  - Epenthesis: tʃɪ.poʊ.tɨ.le
  - /t/ glotalling: tʃɪ.poʊʔ.le

Constraint Based Phonology

*tl seems like a *constraint* you know about your language, and there are multiple ways you can try to repair violations of that constraint.

Constraint based phonology tries to ask: What if all of phonology was actually this way?

Commonly Recurring Patterns

Both English and Somali have rules that do something to two coronal stops in a sequence.

English:

EPENTHESIS: $\varnothing \rightarrow V \,/\, \begin{bmatrix} \text{COR} \\ -\text{cont} \\ -\text{son} \end{bmatrix} ____ \begin{bmatrix} \text{COR} \\ -\text{cont} \\ -\text{son} \end{bmatrix}$

UR	weɪt+d
EPENTHESIS	weɪtid
…	…
SR	weɪɾɨd

Somali:

DELETION: $\begin{bmatrix} \text{COR} \\ -\text{cont} \\ -\text{son} \end{bmatrix} \rightarrow \varnothing \,/\, \begin{bmatrix} \text{COR} \\ -\text{cont} \\ -\text{son} \end{bmatrix} ____$

UR	bad+ta
DELETION	bada
…	…
SR	bada

Commonly Recurring Patterns

Instead of having two completely different rules for these two language, what if we instead said there's a problem they have in common (two coronal stops in a sequence), and they're choosing different strategies to repair it?

Optimality Theory

Optimality Theory (OT) tries to understand phonology as a set of competing constraints.

Markedness Constraints

Constraints that say "Don't do something"

- *TL: Don't have a [tl] or [dl] onset

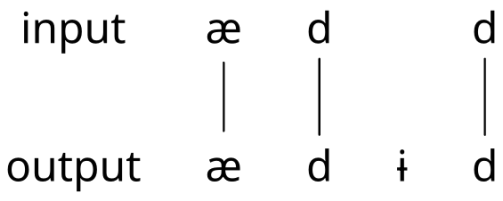
Faithfulness Constraints

Constraints that say “Don’t change anything”

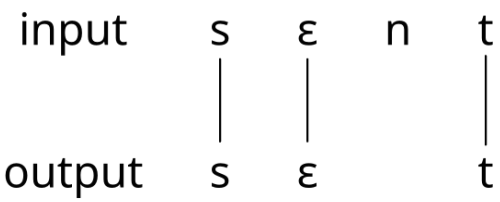
- Ident: Don’t change any features
- Max: Every segment in the input should have a corresponding segment in the output.
- Dep: Every segment in the output should have a corresponding segment in the input.

Faithfulness Constraints

Violates DEP



Violates Max



Optimality Theory

A recurring pattern across languages is that nasal + voiceless sequences are dispreferred.

*NT

Optimality Theory

What are all the ways we could satisfy *NT for this input?

sɛnt

Optimality Theory

Input: sɛnt

- Epenthesis: sɛnɨt
- C_1 Deletion: sɛt
- C_2 deletion: sɛn
- Voicing: sɛnd
- Denasalization: sɛtt

Optimality Theory

If we are working with a language that chooses Epenthesis, we need to ask, why didn’t it choose any of the other repair options?

Optimality Theory

Input: sɛnt

- Epenthesis: sɛnɨt
  - Violates Dep
- C_1 Deletion: sɛt
  - Violates Max
- C_2 deletion: sɛn
  - Violates Max
- Voicing: sɛnd
  - Violates Ident-Voice
- Denasalization: sɛtt
  - Violates Ident-Nasal

Optimality Theory

If a language has epenthesis between nasals and voiceless obstruents:

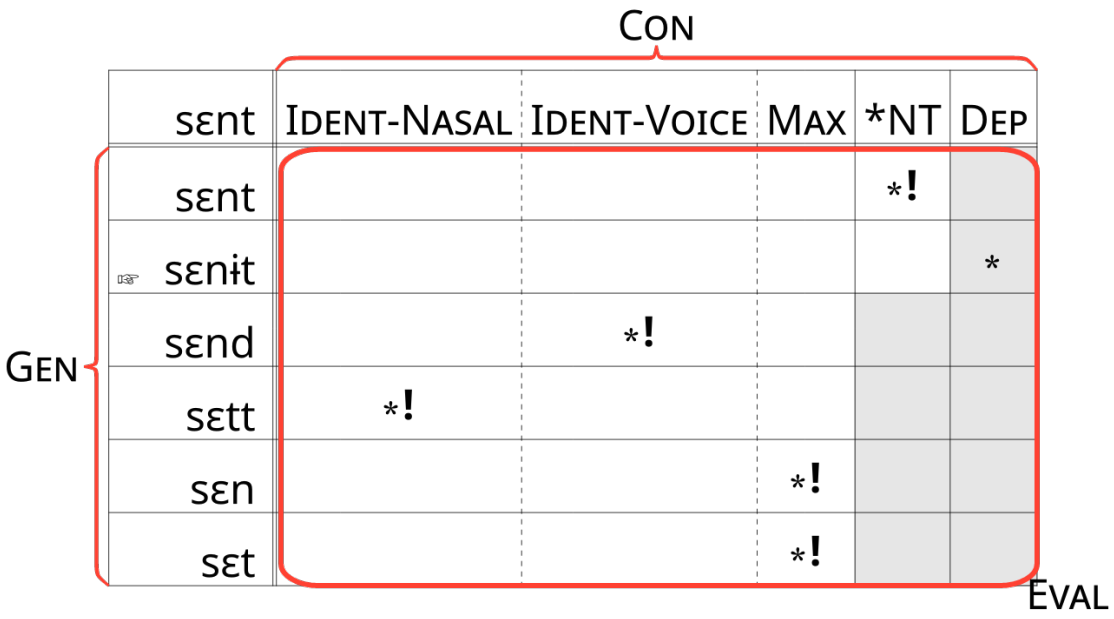
- Avoiding *NT is more important than satisfying Dep
- Satisfying Max, Ident-Voice, Ident-Nasal are all more important than avoiding *NT

Optimality Theory

An OT Tableau

sɛnt	IDENT-NASAL	IDENT-VOICE	MAX	*NT	DEP
sɛnt				*!	
☞ sɛnɨt					*
sɛnd		*!			
sɛtt	*!				
sɛn			*!		
sɛt			*!		

Optimality Theory



Optimality Theory

- Con: The set of constraints
- Gen: Generator of a potentially infinite number of output forms given the input form.
- Eval: The Evaluator that scores each Input:Output pair

Optimality Theory

The fact that there are so many recurring patterns across languages means maybe there's a universal set of constraints.

Optimality Theory

Example: Yoruba

- ri+igba ? rigba
- ri+aʃɔ ? raʃɔ
- ri+ɔbɛ ? rɔbɛ

Constaints:

- *VV: No two vowels in squence
- Max
- Dep
- Ident

Optimality Theory

1. List many possible outputs for an input
2. Work through the necessary constraint ranking

Optimality Theory

riaʃɔ	*VV	DEP	MAX
riaʃɔ	*!		
☞ raʃɔ			*
rijaʃɔ		*!	